Java Swing (JFC) and Swing Components

Swing (JFC) – Lab Practice Questions

1. **Student Registration Form**
   Develop a Java Swing application to create a **Student Registration Form** using JLabel, JTextField, JRadioButton, JCheckBox, JComboBox, and JButton.
   - Accept student name, register number, gender, course, and hobbies.
   - Use a suitable layout manager.
   - Display the entered details when the **Submit** button is clicked.
   - Provide a **Clear** button to reset all fields.
2. **Simple Calculator using Swing Components**
   Develop a GUI-based calculator using Swing components such as JTextField and JButton.
   - Provide buttons for addition, subtraction, multiplication, and division.
   - Accept two numbers from the user.
   - Display the result in a text field or label.
   - Handle invalid input and division by zero using appropriate exception handling.
3. **Student Mark List Application**
   Create a Swing application to accept the **name, register number, and marks in three subjects** using text fields.
   - Calculate total, average, and grade.
   - Display the result using JLabel or JTextArea.
   - Use JButton for **Calculate**, **Clear**, and **Exit** operations.
   - Validate that the entered marks are within the range 0–100.
4. **Login and User Authentication Interface**
   Design a Java Swing application containing JLabel, JTextField, JPasswordField, and JButton components to create a **Login Form**.
   - Accept username and password.
   - Verify the entered credentials against predefined values.
   - Display appropriate success or error messages using JOptionPane.
   - Include **Reset** and **Exit** buttons.
   - Ensure that the password is not displayed as plain text.
5. **Library Book Management GUI**
   Develop a Swing-based **Library Book Management** interface using JTextField, JComboBox, JTable, JButton, JOptionPane, and appropriate layout managers.
   The application should allow the user to:

- Enter book ID, title, author, and category.
- Add book details to a table.
- Delete a selected book from the table.
- Clear the input fields.
- Display a message when no table row is selected for deletion.